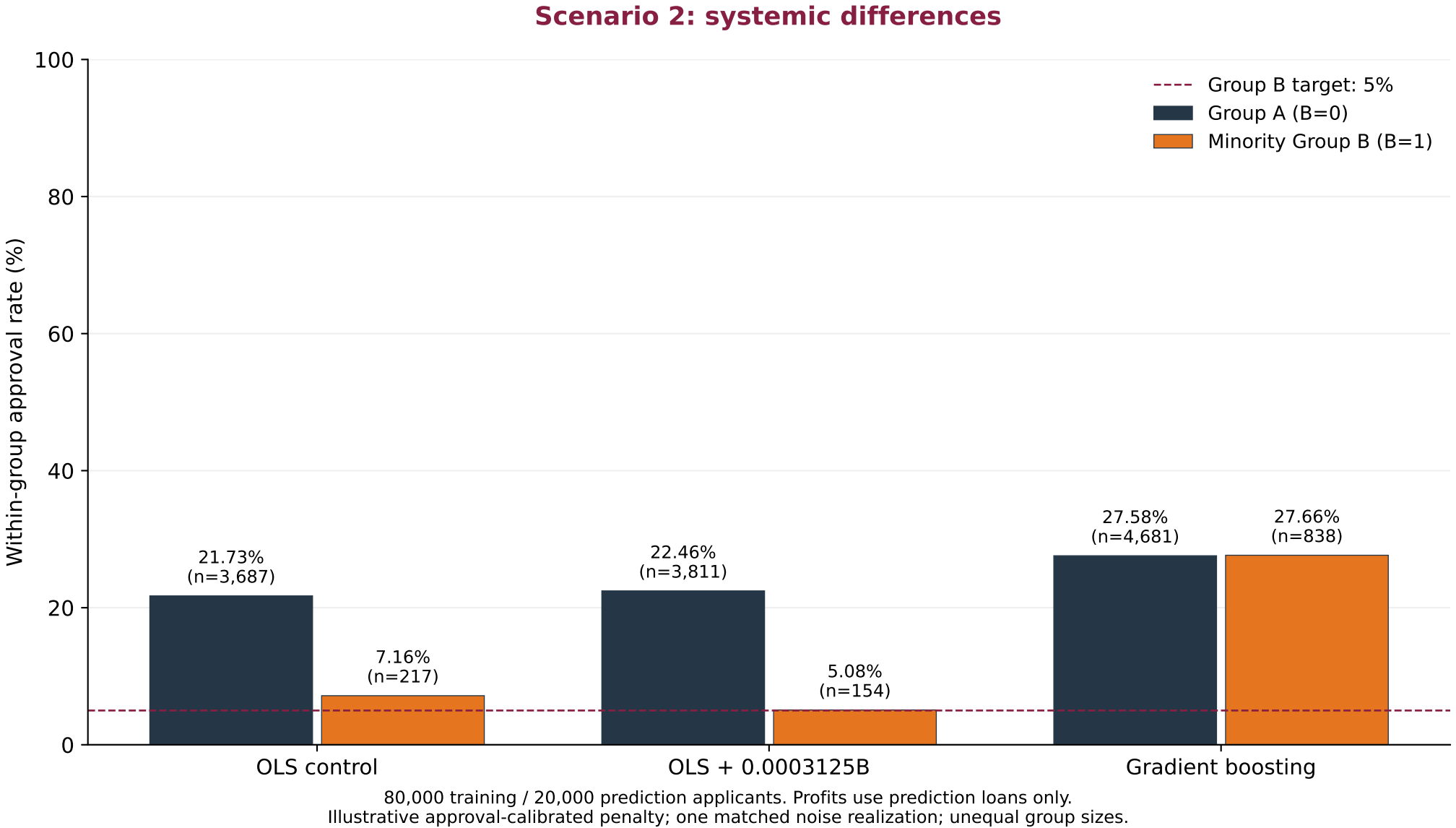
Scenario 2: systemic differences
Group B target: 5%
Group A (B=0)
Minority Group B (B=1)
Within-group approval rate (%)
100
80
60
40
20
0
21.73%
(n=3,687)
7.16%
(n=217)
22.46%
(n=3,811)
5.08%
(n=154)
27.58%
(n=4,681)
27.66%
(n=838)
OLS control
OLS + 0.0003125B
Gradient boosting
80,000 training / 20,000 prediction applicants. Profits use prediction loans only.
Illustrative approval-calibrated penalty; one matched noise realization; unequal group sizes.